

A PROJECT REPORT ON

EXPLORING ACTIVATION FUNCTIONS (ReLU, Sigmoid) FOR MNIST IMAGE CLASSIFICATION USING DEEP LEARNING

Submitted in partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

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ABSTRACT

Image classification is an important application of deep learning, where neural networks learn patterns from image data to classify them into predefined categories. This project presents a comparative study of Sigmoid and ReLU (Rectified Linear Unit) activation functions for handwritten digit image classification using the MNIST dataset. The dataset contains grayscale images of handwritten digits from 0 to 9, which are normalized and transformed into 784-dimensional input features.

Two fully connected Artificial Neural Network (ANN) models with the same 784–128–64–10 architecture are developed using TensorFlow and Keras. The first model uses the Sigmoid activation function in its hidden layers, while the second uses ReLU, with Softmax used in the output layer for multi-class classification. Both models are trained using the same experimental settings to provide a fair comparison.

The models are evaluated using accuracy, loss, confusion matrix, precision, recall, and F1-score. The Sigmoid model achieved a test accuracy of 97.33% with a test loss of 0.08598, while the ReLU model achieved 97.70% accuracy with a lower test loss of 0.07862. The Sigmoid and ReLU models produced 267 and 230 misclassified images, respectively. These results provide a practical comparison of activation functions and demonstrate their influence on the learning and classification performance of neural networks.

Keywords: Deep Learning, Artificial Neural Networks, Image Classification, MNIST, Activation Functions, Sigmoid, ReLU, Softmax, TensorFlow, Keras, Handwritten Digit Recognition.

Table of Contents

1. PROBLEM DEFINITION	6
1.1 Background and Content	6
1.2 Problem Statement	6
1.3 Objectives of the Project	7
1.4 Scope and Organization of the Report	7
2. LITERATURE REVIEW	9
2.1 Introduction	9
2.2 Foundations and Previous Research	9
2.3 Comparison of Existing Methods	10
3. Methodology	11
3.1 Dataset Profiling and Preprocessing Architecture	11
3.2 Data Transformation and Preparation	12
3.3 Applied Neural Networks	12
3.4 Comprehensive Technology Stack and Infrastructure	13
4. RESULTS AND DISCUSSION	18
4.1 Model Performance Evaluation and Metrics	18
4.2 Model Discrimination and Error Analysis	21
4.3 Comparative Analysis of Sigmoid and ReLU	21
5. CONCLUSION AND FUTURE SCOPE	22
5.1 Summary of Project Achievements	22
5.2 Limitations	22
5.3 Future Scope	22

Table of Images

1.PROBLEM DEFINITION	5
1.1 Samples of MNIST images (28 X 28 pixels).	7
3. Methodology	11
3.1 Training Accuracy of Sigmoid vs ReLU	13
3.2 Validation Accuracy of Sigmoid vs ReLU	14
3.3 Training Loss of Sigmoid vs ReLU	14
3.4 Validation Loss of Sigmoid vs ReLU	15
3.5 Confusion Matrix - Sigmoid Model	15
3.6 Confusion Matrix of ReLU Model	16
3.7 Classification Report of Sigmoid (F1, Recall, Precision, Support)	16
3.8 Classification Report of ReLU (F1, Recall, Precision, Support)	17
4. RESULTS AND DISCUSSION	18
4.1 Final Test Accuracy Comparison (Sigmoid vs ReLU)	18
4.2Total Correctly Classified by ReLU (9770)	19
4.3 Total Misclassified by ReLU (230)	19
4.4 Total Correctly Classified by Sigmoid (9733)	20
4.5 Total Misclassified by Sigmoid (267)	20

1. PROBLEM DEFINITION

1.1 Background and Content

Image classification is an area where AI and Deep Learning is applied extensively in terms of learning from the images and classifying them into certain categories. Artificial Neural Networks (ANNs) have been used widely for the purpose of image classification since they automatically learn the relationship between features using various layers of connected networks.

The performance of a neural network largely depends on the selection of an activation function that brings about non-linearity and allows the network to learn complex relationships. Sigmoid and Rectified Linear Unit (ReLU) are two types of activation functions having various mathematical and learning properties.

MNIST data set is a well-known collection of grayscale images containing handwritten digits from 0 to 9 and has been used widely for the assessment of image classification models. The current project is aimed at comparing Sigmoid and ReLU activation functions through a common neural network model.

1.2 Problem Statement

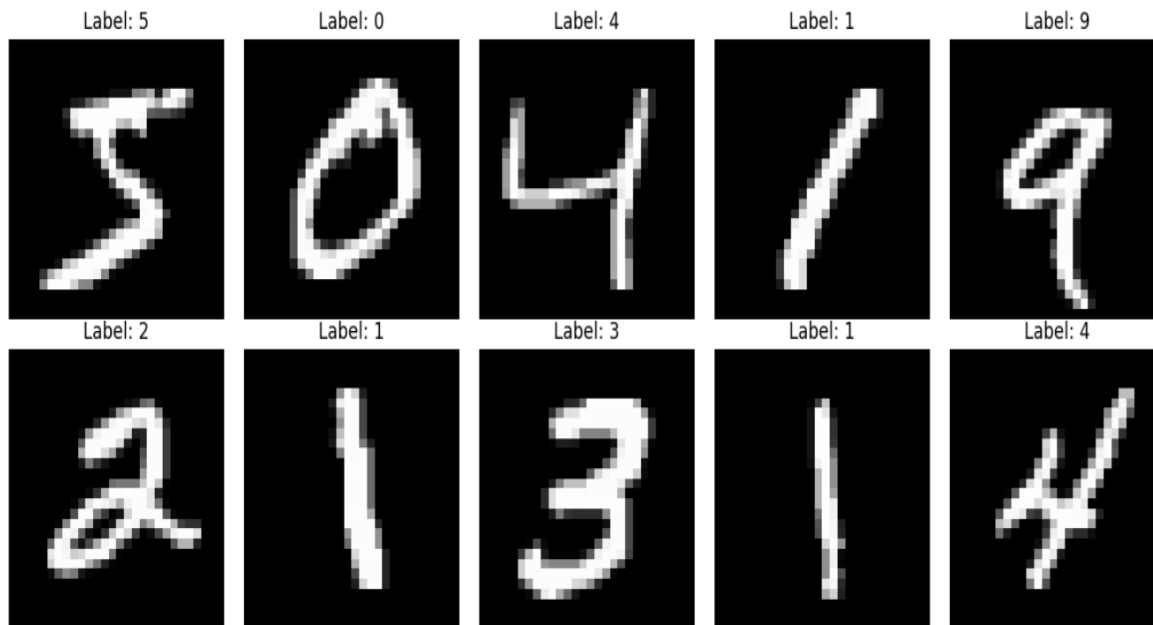
Several factors contribute to determining the performance of a neural network such as the network architecture, the learning process, and the activation function used. Sigmoid, although helpful in adding nonlinearity, causes issues in the convergence of the model due to the small gradient value for some inputs. On the other hand, the ReLU function provides a simple activation approach but has different learning properties compared to the sigmoid function.

The problem that is to be addressed in this project involves evaluating the Sigmoid and ReLU activation functions in the same neural network architecture for MNIST handwritten digit classification. The main objective is to conduct a comparative study in terms of accuracy, losses, and prediction error using the two activation functions on a fixed architecture and experiment settings.

1.3 Objectives of the Project

The primary objectives of this project are to systematically study and compare the performance of different activation functions in a neural network-based image classification task. The objectives of this study include:

- 1. To preprocess and normalize the MNIST handwritten digit dataset and transform the images into suitable input features for the neural network.
- 2. To design and implement a fully connected Artificial Neural Network (ANN) with the architecture 784–128–64–10 using TensorFlow and Keras.
- 3. To develop two models using Sigmoid and ReLU activation functions in the hidden layers while maintaining the same network architecture and experimental settings.
- 4. To train and evaluate both models using accuracy and loss, along with confusion matrices and classification metrics such as precision, recall, and F1-score.
- 5. To compare the learning behavior, prediction errors, and classification performance of the Sigmoid and ReLU models and analyze the effect of activation-function selection on MNIST image classification.



1.1 Samples of MNIST images (28 X 28 pixels).

1.4 Scope and Organization of the Report

The scope of the project includes the design and analysis of deep learning based handwritten digits classifier using MNIST dataset. The project considers data pre-processing, neural network design, and the comparative analysis of Sigmoid and ReLU activation functions by applying the same network architecture and

experimental environment. The efficiency of both classifiers is measured by accuracy, loss, confusion matrix, precision, recall, and F1 score.

The structure of the report goes as follows: In Section 2, the relevance of the project and significance of activation functions in deep learning are discussed. In Section 3, the methodology of the project including dataset, pre-processing techniques, neural network architecture, activation functions and technologies used are described. Section 4 deals with experimental results and comparison of Sigmoid and ReLU models. Lastly, Section 5 is for conclusion, limitations, and future scope.

2. LITERATURE REVIEW

2.1 Introduction

Deep learning is a widely adopted method for image classification since neural networks allow automatic discovery of useful patterns in image data. Artificial Neural Network (ANN) is a collection of neurons organized into layers, where each layer performs transformations on the data using mathematical equations.

One of the important elements of neural networks is the activation function, which allows adding non-linearity and making the neural network capable of learning complex patterns. There are different activation functions with specific mathematical characteristics, which can affect the learning process of neural networks.

In this project, performance of two most popular activation functions – Sigmoid and ReLU – is investigated for handwritten digit classification using MNIST dataset. This review of literature forms the basis for understanding neural networks, activation functions, and their application for image classification.

2.2 Foundations and Previous Research

Artificial Neural Networks mimic the elementary workings of biological neurons and are used in a variety of applications, primarily related to classification and pattern recognition. A neural network is made up of an input layer, one or more hidden layers, and an output layer. Each neuron takes a linear combination of its inputs and passes through the activation function to generate its output.

The Sigmoid activation function is responsible for mapping the input data into a range of 0 to 1. It has been historically employed in neural networks due to the differentiability of the function. For large values of positive or negative input values, the gradient of the function tends to be very small, which can affect the learning process adversely.

The Rectified Linear Units activation function outputs zero for negative input values and preserves the positive values. Due to easy calculations and the provision of steeper gradients for the positive inputs, ReLU has become quite popular among neural networks.

Thus, different activation functions have been explored in previous research of neural networks in order to enhance learning efficiency and classification. The current project explores the difference between Sigmoid and ReLU activation functions within the same neural network framework.

2.3 Comparison of Existing Methods

Different activation functions can result in different learning patterns even when the architecture of the neural network used stays the same. The sigmoid function returns output in the range of 0 to 1, while ReLU gives a linear output for positive values and zero output for negative values.

In the case of image classification tasks, the selection of an activation function can impact the learning pattern in terms of convergence and loss reduction in addition to classification performance. In this research work, instead of making changes in many parameters of a neural network, only the activation function is being changed.

The two approaches investigated in this project are:

Model	Hidden-Layer Activation	Output Activation
Sigmoid Model	Sigmoid	Softmax
ReLU Model	ReLU	Softmax

3. Methodology

The methodology of the research includes multiple steps. First of all, the MNIST dataset is loaded and examined. Then, the normalization of the pixel values from the range 0–255 to 0–1 and the flattening of the 28×28 images into the 784-dimensional input vectors are performed.

Two neural networks that have identical architecture – 784-128-64-10 are built. In the case of the first network, the activation functions for hidden layers are Sigmoid, while in the case of the second one – ReLU. The activation function for the output layer is Softmax.

Both networks are trained under identical experimental conditions and tested against the separate MNIST test dataset.

3.1 Dataset Profiling and Preprocessing Architecture

Empirical groundings of the project lie in the usage of the MNIST handwritten digit database – a well-known dataset for image classification that includes 70,000 grayscale images that represent handwritten digits from 0 to 9. Each image has a resolution of 28×28 pixels, which makes 784 pixel values per image. In total, the dataset includes 60,000 training images and 10,000 test images.

Data Component	Data Type	Representative Attributes / Metrics
Image Characteristics	Grayscale Image	28×28 pixel resolution, 70,000 total images
Pixel Values	Numeric / Integer	Pixel intensity range: 0–255
Target Classes	Categorical	Digits 0–9, 10 classification classes
Training Dataset	Image / Label Pairs	60,000 labeled images
Testing Dataset	Image / Label Pairs	10,000 labeled images

Initially, pixel values are between 0 and 255. In order to use them as a proper numerical representation in the neural network, the values are normalized to the range between 0 and 1. After that, 28×28 image matrices are transformed to one-dimensional vectors that include 784 input features. Initial training set is split into two – training and validation datasets with 54,000 and 6,000 images respectively.

3.2 Data Transformation and Preparation

During the preprocessing phase, the raw images of the MNIST dataset are converted to a form acceptable as an input to the fully connected neural networks. This is done through the process of pixel normalization where pixel intensities are divided by 255 to change the range of values from 0–255 to 0–1.

This is due to the fact that the neural network model used in this project has fully connected Dense layers. Therefore, 28×28 images are transformed into 784-dimensional vectors. Corresponding digit labels remain unchanged and are in the range of integers from 0 to 9.

3.3 Applied Neural Network Models

To establish a controlled comparative experiment, two fully connected Artificial Neural Network architectures are implemented using the same input dimensions, hidden-layer sizes, output layer, and training configuration. The primary difference between the two models is the activation function used in their hidden layers.

1. Sigmoid-Based Artificial Neural Network:

The first model consists of an input layer accepting 784 features, followed by two hidden layers containing 128 and 64 neurons, respectively. The Sigmoid activation function is applied to both hidden layers. The final output layer contains 10 neurons and uses the Softmax activation function to generate class probabilities for the ten handwritten digit categories.

The model is trained using the Adam optimizer with sparse categorical cross-entropy as the loss function. A batch size of 128 is used and the model is trained for 10 epochs.

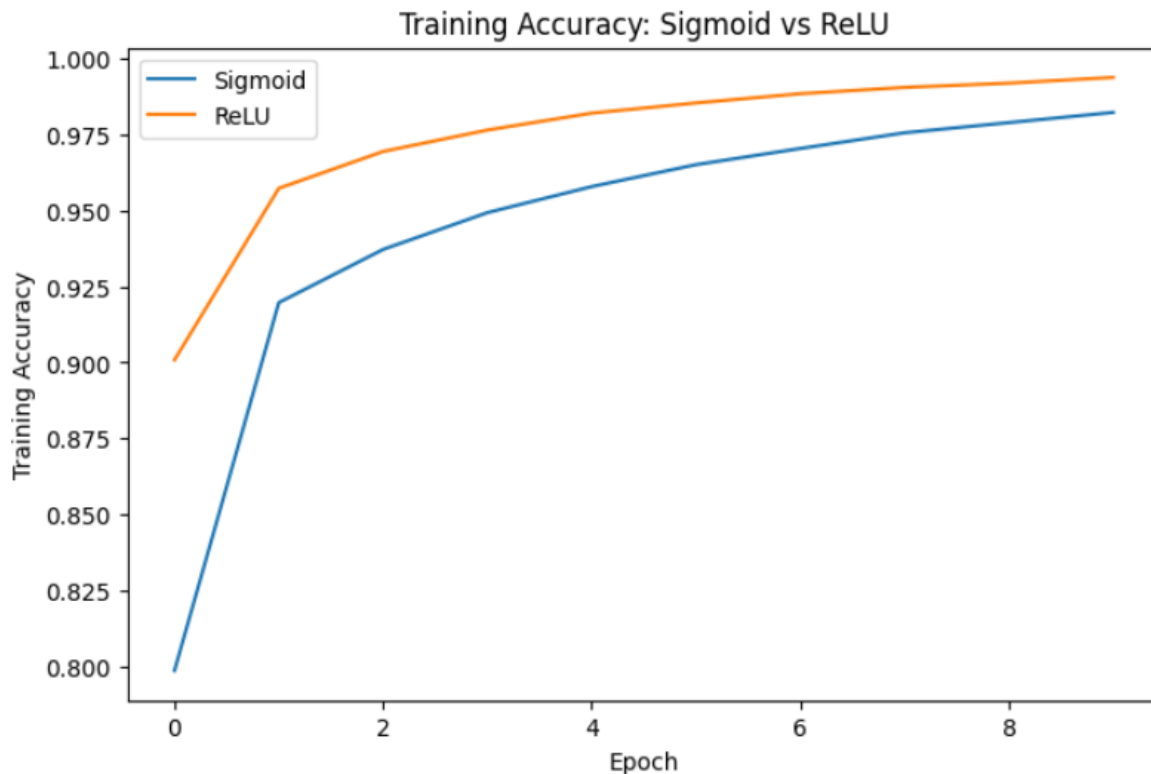
2. ReLU-Based Artificial Neural Network:

The second model uses the same 784–128–64–10 architecture to maintain a controlled comparison. The two hidden layers contain 128 and 64 neurons and use the ReLU activation function. The output layer consists of 10 neurons with Softmax activation for multi-class classification.

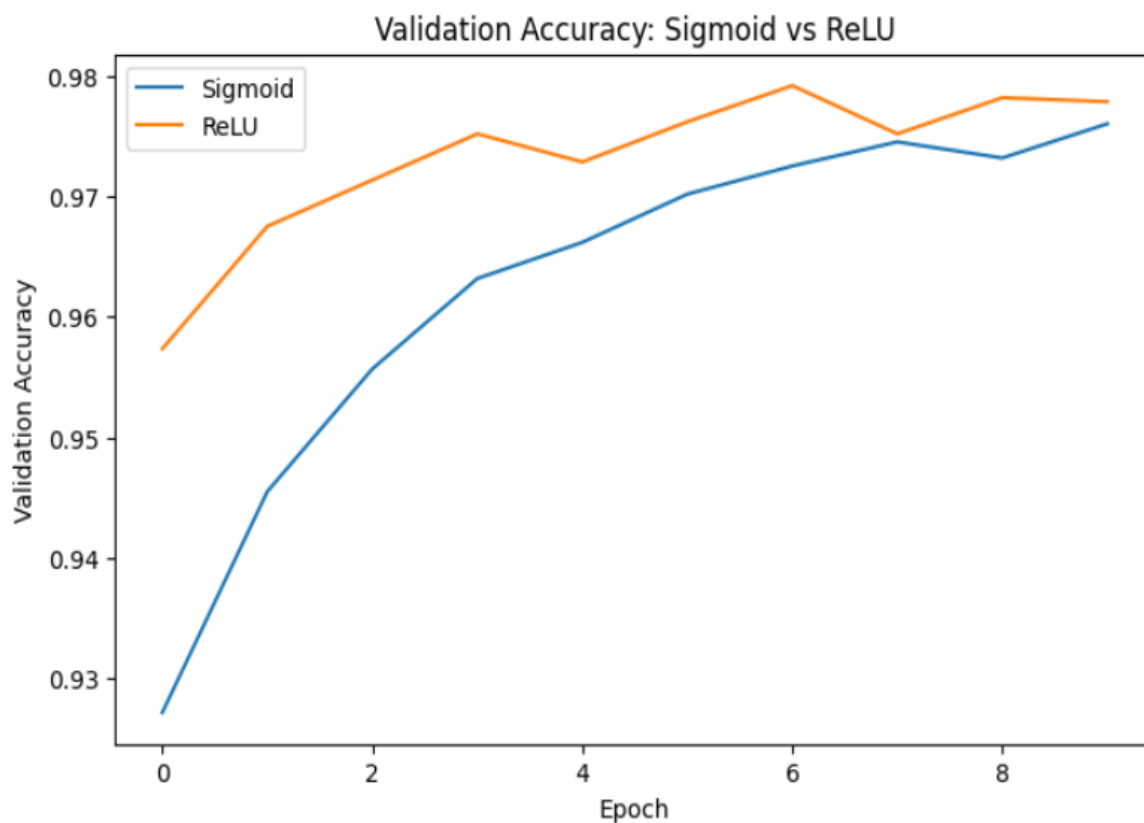
The same Adam optimizer, sparse categorical cross-entropy loss function, batch size of 128, and 10 training epochs are used. Maintaining identical experimental settings allows the observed differences in model behavior to be examined primarily in relation to the hidden-layer activation function.

3.4 Comprehensive Technology Stack and Infrastructure

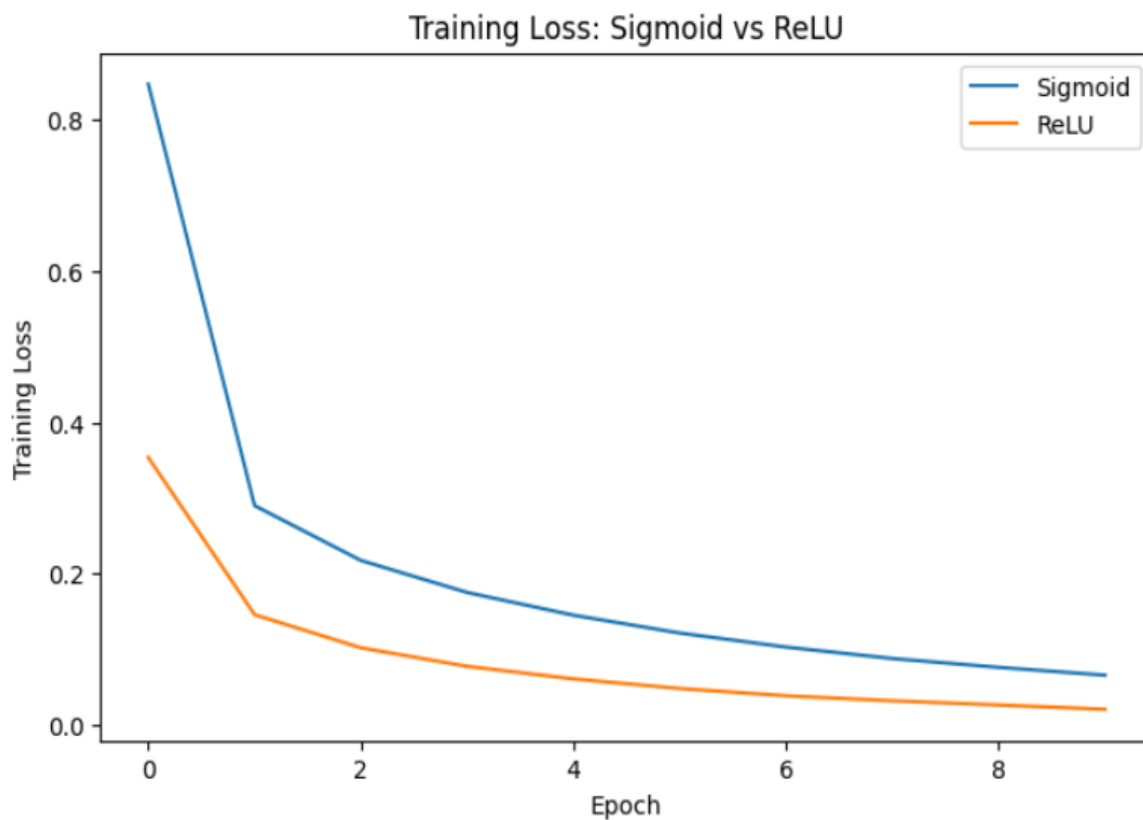
- **Programming Language & Core Environment:** Python served as the primary programming language for implementing the complete experimental workflow due to its extensive ecosystem for numerical computing, data analysis, visualization, and deep learning applications.
- **Numerical Computing & Data Handling:** NumPy was utilized for numerical operations, array manipulation, image reshaping, and transformation of the MNIST image data. Pandas was used for structured data handling and organization of experimental results.
- **Deep Learning Frameworks:** TensorFlow with its Keras API was used to construct, compile, and train the Artificial Neural Network models. The framework provided the required Dense layers, activation functions, optimization procedures, loss functions, and training mechanisms for implementing the Sigmoid and ReLU models.
- **Data Visualization & Analysis:** Matplotlib and Seaborn were used to visualize sample MNIST images, training and validation performance, loss curves, confusion matrices, and other graphical results required for comparative analysis.



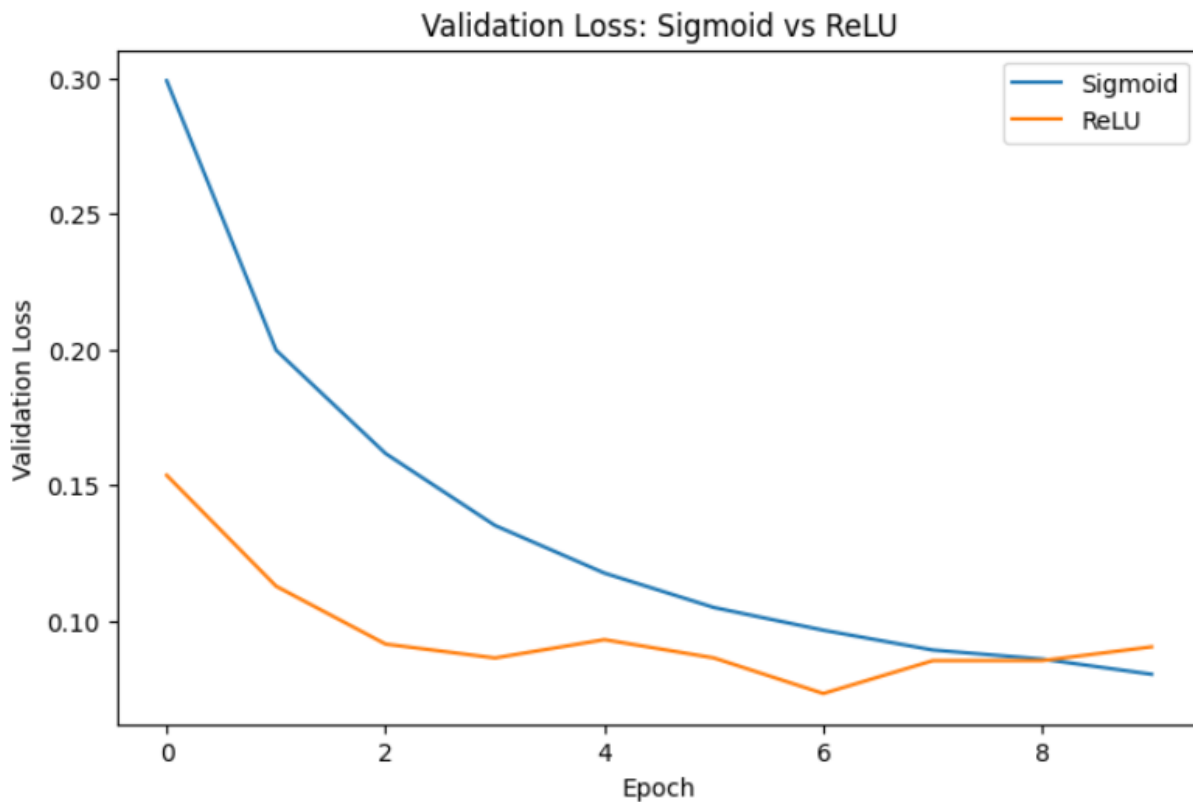
3.1 Training Accuracy of Sigmoid vs ReLU



3.2 Validation Accuracy of Sigmoid vs ReLU

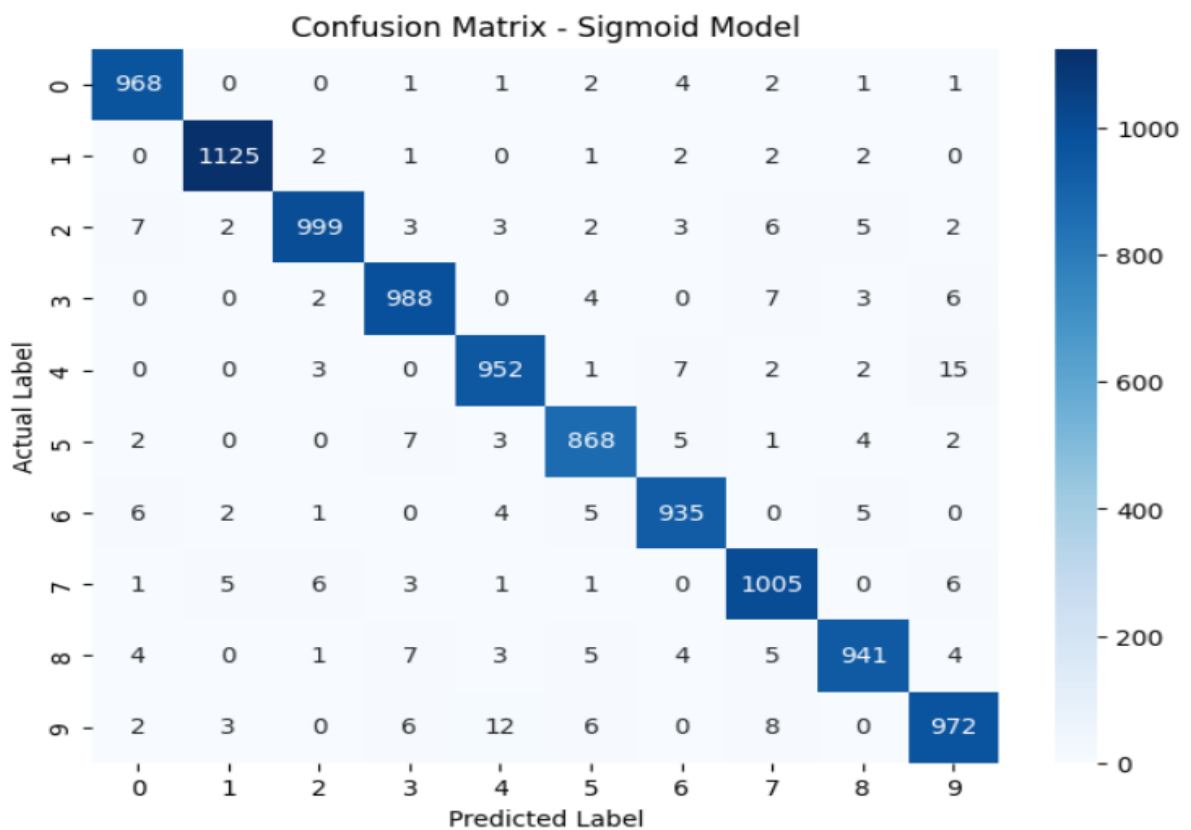


3.3 Training Loss of Sigmoid vs ReLU

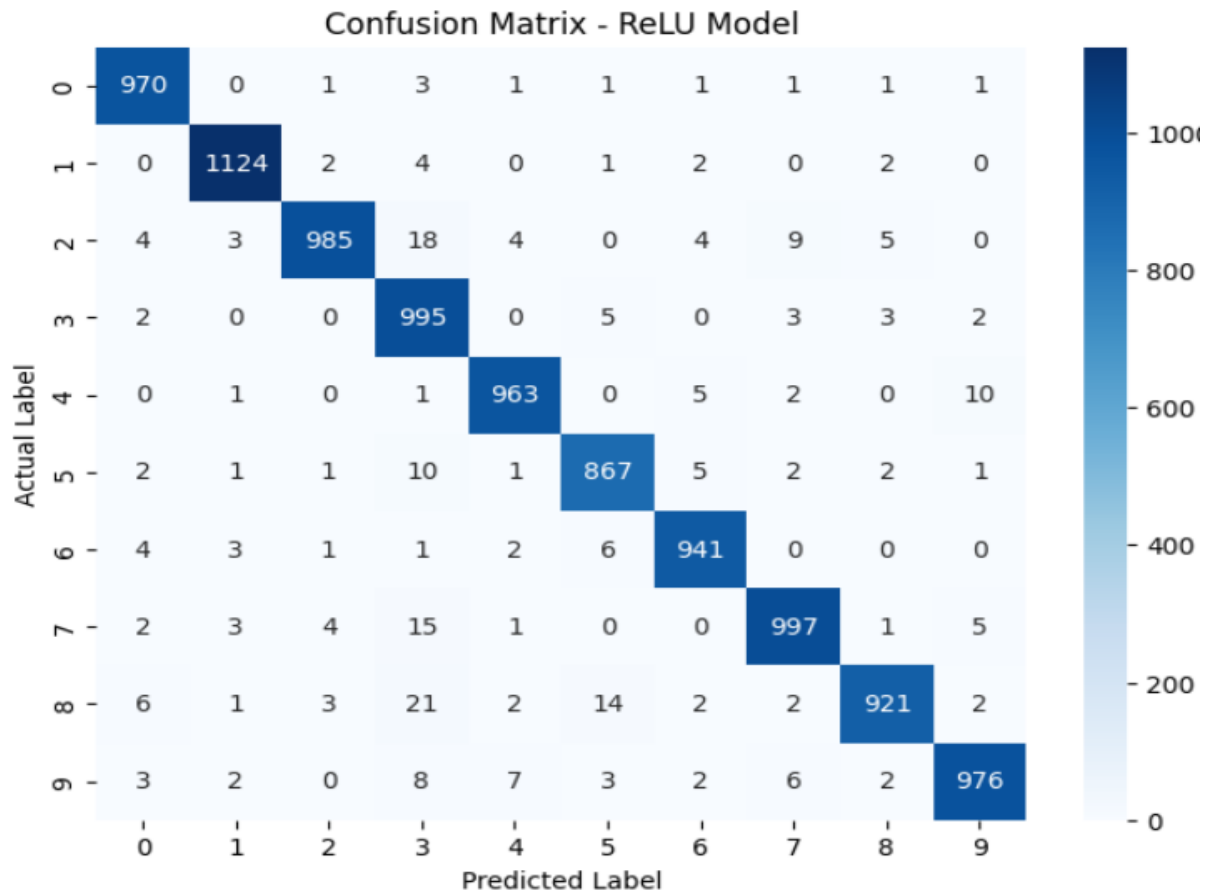


3.4 Validation Loss of Sigmoid vs ReLU

- Model Evaluation: Scikit-learn was utilized for generating confusion matrices and classification reports, including precision, recall, and F1-score, to evaluate the classification performance of the trained models.



3.5 Confusion Matrix - Sigmoid Model



3.6 Confusion Matrix of ReLU Model

Classification Report - Sigmoid Model

	precision	recall	f1-score	support
0	0.98	0.99	0.98	980
1	0.99	0.99	0.99	1135
2	0.99	0.97	0.98	1032
3	0.97	0.98	0.98	1010
4	0.97	0.97	0.97	982
5	0.97	0.97	0.97	892
6	0.97	0.98	0.97	958
7	0.97	0.98	0.97	1028
8	0.98	0.97	0.97	974
9	0.96	0.96	0.96	1009
accuracy			0.98	10000
macro avg	0.98	0.98	0.98	10000
weighted avg	0.98	0.98	0.98	10000

3.7 Classification Report of Sigmoid (F1, Recall, Precision, Support)

Classification Report - ReLU Model

	precision	recall	f1-score	support
0	0.98	0.99	0.98	980
1	0.99	0.99	0.99	1135
2	0.99	0.95	0.97	1032
3	0.92	0.99	0.95	1010
4	0.98	0.98	0.98	982
5	0.97	0.97	0.97	892
6	0.98	0.98	0.98	958
7	0.98	0.97	0.97	1028
8	0.98	0.95	0.96	974
9	0.98	0.97	0.97	1009
accuracy			0.97	10000
macro avg	0.97	0.97	0.97	10000
weighted avg	0.97	0.97	0.97	10000

3.8 Classification Report of ReLU (F1, Recall, Precision, Support)

- **Computational Environment:** Google Colab was used as the primary development and execution environment for loading the MNIST dataset, training both neural network models, generating predictions, and performing the final experimental analysis.

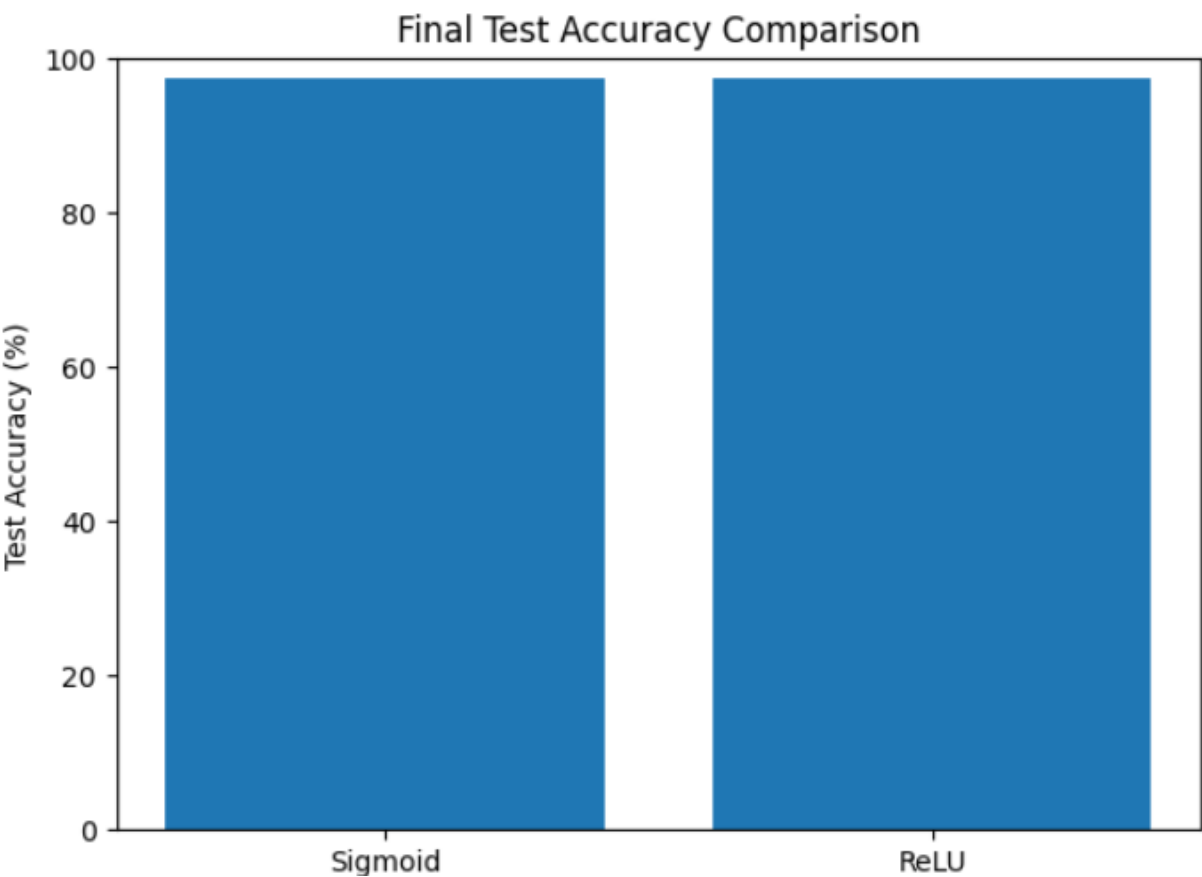
Infrastructure Layer	Technology / Framework Selected
Computational Engine	Python 3.13.15, Google Colab
Deep Learning Model	TensorFlow 2.20.0, Keras
Algorithmic & Analysis Libraries	NumPy 2.1.3, Pandas 2.2.3, Scikit-learn 1.6.1, Matplotlib 3.10.0, Seaborn 0.13.2

4. RESULTS AND DISCUSSION

4.1 Model Performance Evaluation and Metrics

Predictive performance of the trained models was examined by applying the held-out MNIST test data set with 10,000 images of handwritten digits. Performance of both the models was analyzed based on different metrics including accuracy, test loss, and number of misclassification. As it is already known that the main focus of this project is comparison between performance of the Sigmoid and ReLU activation functions, so, the test data and architecture used for both the models is the same.

Model Architecture	Test Accuracy	Test Loss	Misclassified Images
Sigmoid-Based ANN	0.9733	0.08598	267
ReLU-Based ANN	0.9770	0.07862	230

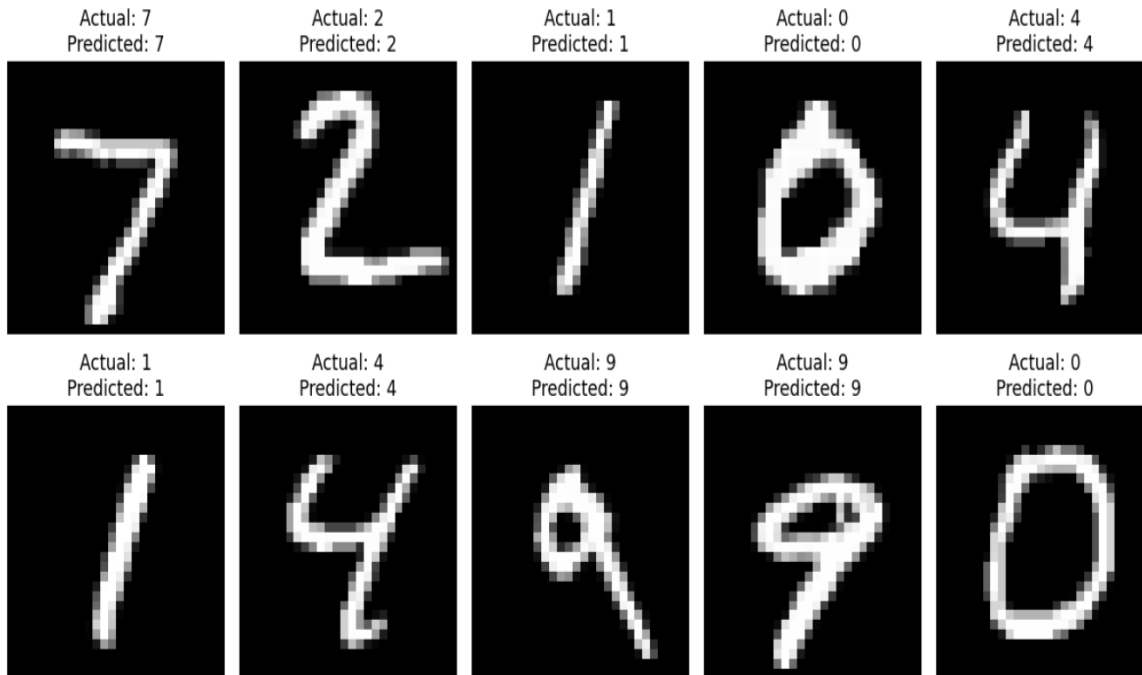


4.1 Final Test Accuracy Comparison (Sigmoid vs ReLU)

From the experimental outcomes, it can be noted that the Sigmoid-Based ANN gave a test accuracy of 97.33% with a test loss of 0.08598 with 267 misclassified images. On the other hand, the ReLU-Based ANN provided a test accuracy of 97.70% with a reduced test loss of 0.07862 with 230 misclassified images. The

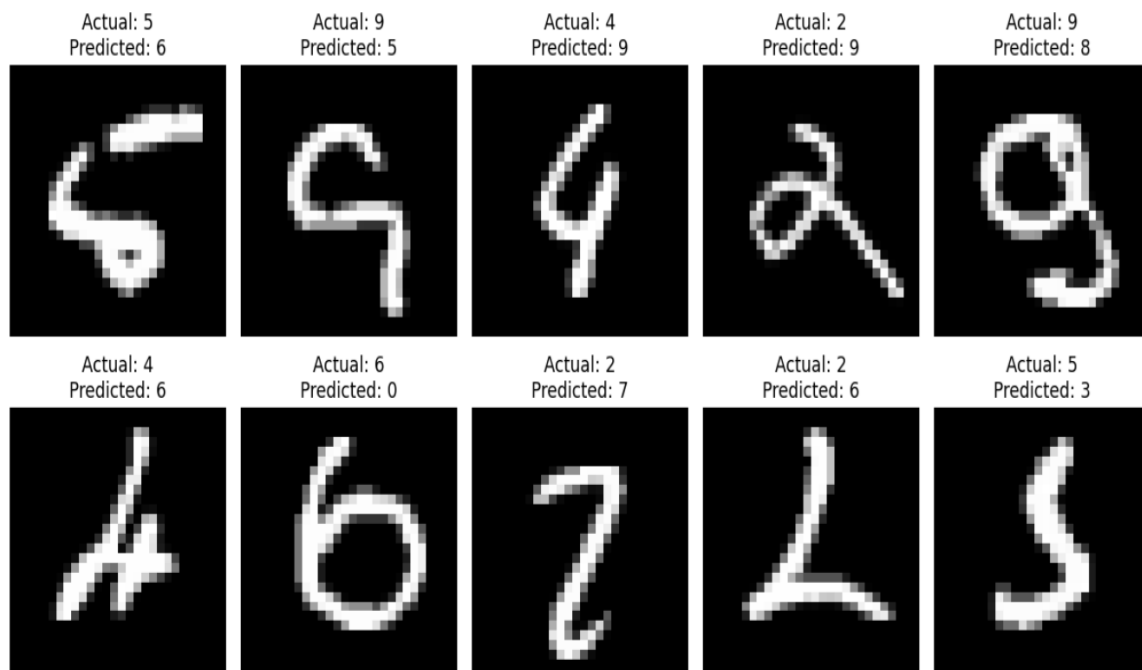
outcome obtained from the experiment can show that in the experimental set-up adopted for this project, the ReLU-based model resulted in a test accuracy that was 0.37 percentage point higher with 37 fewer misclassifications as compared to the Sigmoid-based model. The reduction in the test loss in the ReLU model also suggests a difference between the prediction power of the two models when applied to the unseen test data.

Total correctly classified: 9770



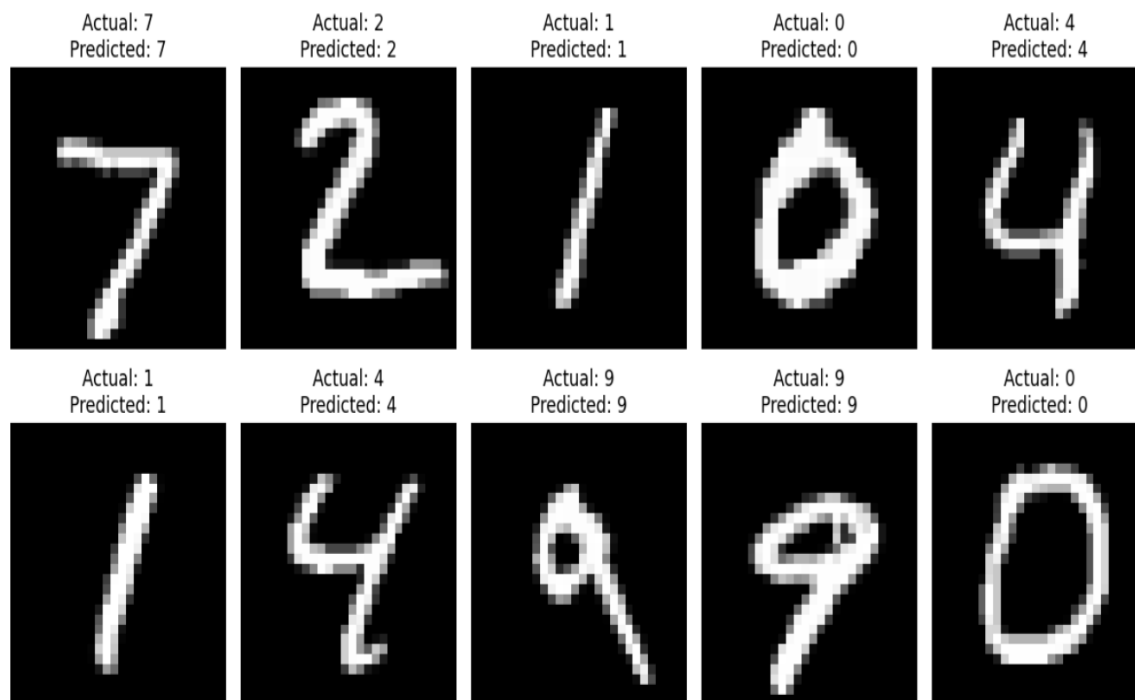
4.2 Total Correctly Classified by ReLU (9770)

Number of misclassified images: 230



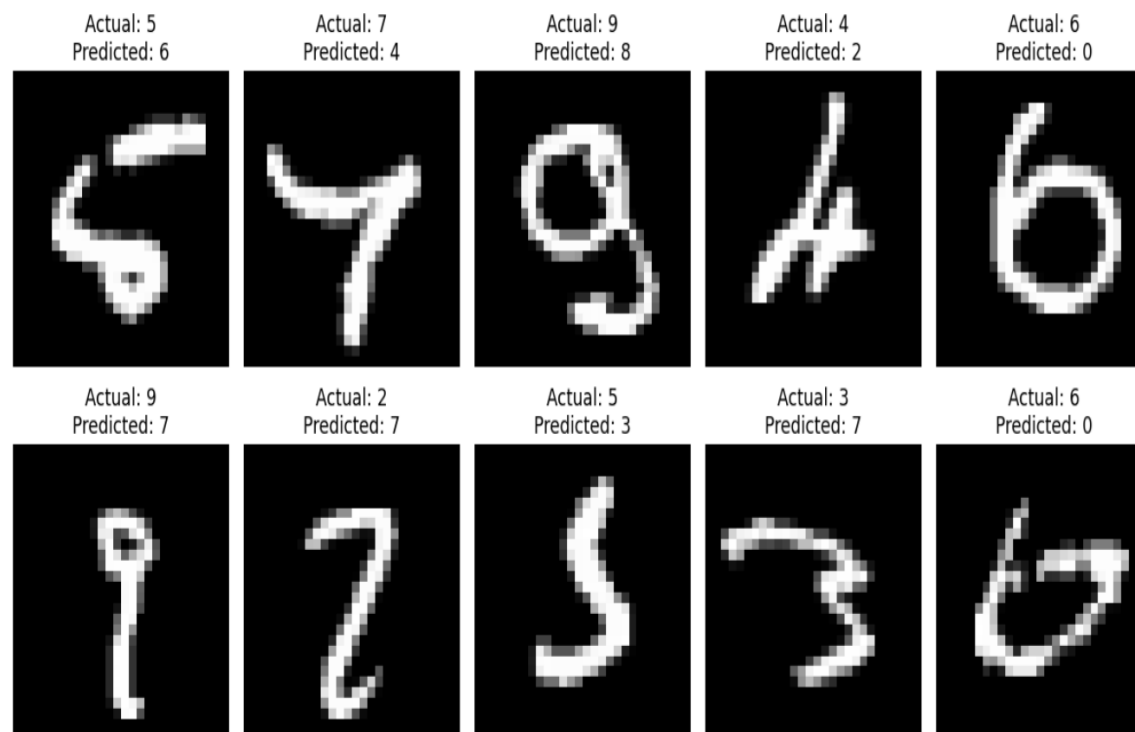
4.3 Total Misclassified by ReLU (230)

Total correctly classified: 9733



4.4 Total Correctly Classified by Sigmoid (9733)

Number of misclassified images: 267



4.5 Total Misclassified by Sigmoid (267)

4.2 Model Discrimination and Error Analysis

In this project, the error analysis will be done based on the classification errors from the Sigmoid and ReLU models on the test dataset of 10,000 images from the MNIST database. In this case, predictions of the two models were compared with the corresponding actual labels to find out any wrongly classified images and analyze the prediction differences between them.

Specifically, the Sigmoid-based neural network made 267 classification errors on the test set, while the ReLU-based network made 230 errors. Accordingly, the error rates were approximately 2.67% and 2.30%. The difference shows that under the experimental conditions used in this project, the ReLU network made fewer classification mistakes on unseen test data.

Moreover, the errors have been analyzed through the use of confusion matrices and examination of the misclassified images. With the help of the confusion matrix, it is possible to get information regarding which handwritten digits were properly classified and which digits were confused with each other. Additionally, the examination of the misclassified images can help determine visually similar digits that are difficult to separate by the neural networks.

4.3 Comparative Analysis of Sigmoid and ReLU

In order to conduct a comparative analysis, we have tested two models with the same neural network architecture, the same dataset and the same training configuration. The models include an input layer with 784 features, two hidden layers with 128 and 64 neurons correspondingly, and an output layer with 10 neurons which uses Softmax activation. The only difference is in activation functions: the first model uses Sigmoid activation in hidden layers, and the second one – ReLU activation in hidden layers.

Test accuracy of Sigmoid based model is 97.33% and test accuracy of ReLU based model is 97.70%. Therefore, ReLU model gives 0.37 percentage points better test accuracy in this experiment. In addition, the ReLU model shows test loss equal to 0.07862 which is smaller than 0.08598 obtained by the Sigmoid model.

In terms of incorrect predictions, Sigmoid model made 267 wrong classifications while ReLU made 230 mistakes. These results show that changing hidden layers' activation function had an impact on classification results of the same neural network architecture. Based on these results, the training and validation curves and confusion matrices will be used to compare learning behavior of both activation functions.

5. CONCLUSION AND FUTURE SCOPE

5.1 Summary of Project Achievements

This project has been successful in implementing and analyzing two neural network models to compare the Sigmoid and ReLU activation functions for MNIST handwritten digit classification. In both cases, the neural networks have the same structure of 784-128-64-10 and the same training setting except the activation function of the hidden layer.

For testing the neural networks, a sample of 10,000 images from MNIST was used. The Sigmoid neural network gave the result of 97.33% in accuracy with a test loss of 0.08598 and 267 misclassified images, whereas the ReLU neural network resulted in 97.70% in accuracy with a test loss of 0.07862 and 230 misclassified images.

5.2 Limitations

Some restrictions inherent to the design of the system must be noted while interpreting its results. First of all, both the models have been designed based on the MNIST data set featuring rather simple gray-scale images of hand-written digits. Moreover, the neural network models employ fully-connected layers with flattened images as input, thus not explicitly taking into account the spatial relationship between the pixels. Furthermore, the project only uses two activation functions (Sigmoid and ReLU) with a given configuration of training parameters.

5.3 Future Scope

The future scope of this project can be enhanced by conducting experiments on other types of activation functions like Leaky ReLU, ELU, and Tanh and then comparing their results with those of the Sigmoid and ReLU activation functions. The current fully connected neural network can also be further extended to CNNs in order to make use of the inherent spatial characteristics of images. Another area which could be considered for enhancing the model is that of hyperparameter tuning, various other architectures, and training configurations. Comparison can also be carried out with more difficult datasets of images to check if the same trend continues.

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